

Question				Marking details	Marks Available				
					AO1	AO2	AO3	Total	Maths Prac
3				Indicative content: The control rods are designed to be lowered into the reactor to absorb more neutrons to reduce or stop nuclear fission, which will reduce the core temperature. The power cut prevents this level of control. Power cut results in failure of pump so coolant cannot circulate. Rods are automatically released fully into the reactor to stop any more fission reactions. Otherwise eventually high enough temperatures to cause a meltdown are reached. There are immediate consequences to human health from a possible explosion. There are long term consequences to health and the environment due to radiation contamination. 5 – 6 marks Detailed description of the effects of the power cut, why T increases and consequences of meltdown. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3 – 4 mark Full description of two of: the effects of the power cut, why T increases and consequences of meltdown. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	6			6	

			1-2 marks Limited description of any of the effects of the power cut, why T increases and consequences of meltdown. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
			Question 3 total	6			6		